RL Unit Quiz

CS 7641: Machine Learning

Solutions

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1 Question 1 - Markov Decision Processes

Part 1. State Transitions & Value Functions (MCMA)

A robot vacuum cleaner navigates an apartment modeled as a Markov Decision Process (MDP). The transition function specifies the probability of moving from one room to another, given an action.

Which of the following statements about MDP state transitions and value functions are true? Select all that apply.

A. **A transition function must assign non-negative probabilities that sum to one for each state-action pair.**
True. Valid transition probabilities must be in [0,1] and sum to 1 for each state-action pair.

B. **The value of a state under a given policy depends on the transition probabilities and the expected rewards.**
True. State values are calculated as expectations over possible next states and rewards.

C. **If transitions are deterministic, the transition probability for each state-action pair is always 1.**
False. It is 1 for the single next state and 0 for all others, but not "always 1" for every possible state.

D. **In an MDP, the sum of transition probabilities across all possible next states can exceed one.**
False. The sum must equal 1, by definition of a probability distribution.

E. **The Bellman expectation equation relates the value of a state to the values of its possible next states.**
True. The Bellman expectation equation uses expected returns under a policy, combining immediate rewards and next state values.

F. **Changing the transition probabilities has no effect on the optimal policy.**
False. Changes in transitions alter the dynamics and can change which actions maximize expected return.

Correct Answers: A, B, E

Part 2. Deterministic vs. Stochastic Scenarios (MCMA)

A reinforcement learning engineer is analyzing different robot tasks to decide whether the environment is deterministic or stochastic.

Which of the following scenarios describe a stochastic environment? Select all that apply.

A. A robot arm moves an object exactly 5 cm to the right every time it receives the "right" command, with no slippage.
Deterministic.

B. A delivery drone tries to land on a rooftop but wind gusts can push it off course, even with the same landing command.
Stochastic.

C. A vacuum robot moves forward until it hits a wall sensor; given the same wall and command, the result is always the same.
Deterministic.

D. A cleaning robot tries to wipe a stain but sometimes the stain partially remains, depending on the cloth's wear.
Stochastic.

E. A warehouse robot places an item on a conveyor belt, which always moves it forward with no jamming or randomness.
Deterministic.

F. A delivery bot navigates icy sidewalks, where slipping can randomly cause it to slide further than commanded.
Stochastic.

Correct Answers: B, D, F

Part 3 – Bellman Update with Nap Reward

Molebert Euler the mole-bot digs through a tunnel of three burrows plus a terminal burrow:

$$B_1 \longrightarrow B_2 \longrightarrow B_3 \longrightarrow B_4 \text{ (terminal)}.$$

Actions in each burrow

- **Dig Forward**: moves to the next burrow, earning worms as a reward.
- **Nap**: stays in the current burrow, earning a small +0.2 worm snack.

Policy

$$\pi(\text{Dig}) = 0.85, \qquad \pi(\text{Nap}) = 0.15.$$

Transition rewards

$$B_1 \to B_2 : +1, \quad B_2 \to B_3 : +3, \quad B_3 \to B_4 : +5.$$

Discount factor and initial values

$$\gamma = 0.75, \qquad V(B_1) = V(B_2) = V(B_3) = V(B_4) = 0.$$

Questions

a) Write the Bellman Expectation Equation for this policy.

b) Compute the updated $V(B_3)$.

c) Compute the updated $V(B_2)$.

d) Compute the updated $V(B_1)$.

e) If Molebert starts in B_1, what is his expected total worm haul after this update step? (Round to two decimal places.)

Solutions

a) **Bellman Expectation Equation**

For any state s,

$$V(s) \;=\; \sum_a \pi(a \mid s) \sum_{s'} P(s' \mid s, a) \left[R(s, a, s') + \gamma V(s') \right].$$

b) **Update for $V(B_3)$**

$$\begin{aligned} V(B_3) &= 0.85\big[5 + 0.75\,V(B_4)\big] + 0.15\big[0.2 + 0.75\,V(B_3)\big] \\ &= 0.85 \times 5 + 0.15 \times (0.2 + 0.75V(B_3)) \\ &= 4.25 + 0.03 + 0.1125\,V(B_3). \\ 0.8875\,V(B_3) &= 4.28 \implies V(B_3) \approx 4.82. \end{aligned}$$

c) **Update for $V(B_2)$**

$$\begin{aligned} V(B_2) &= 0.85\big[3 + 0.75\,(4.82)\big] + 0.15\big[0.2 + 0.75\,V(B_2)\big] \\ &= 0.85 \times 6.615 + 0.03 + 0.1125\,V(B_2). \\ 0.8875\,V(B_2) &= 5.65 \implies V(B_2) \approx 6.36. \end{aligned}$$

d) **Update for $V(B_1)$**

$$\begin{aligned} V(B_1) &= 0.85\big[1 + 0.75\,(6.36)\big] + 0.15\big[0.2 + 0.75\,V(B_1)\big] \\ &= 0.85 \times 5.77 + 0.03 + 0.1125\,V(B_1). \\ 0.8875\,V(B_1) &= 4.93 \implies V(B_1) \approx 5.56. \end{aligned}$$

e) **Expected total worms starting from B_1**

$$\boxed{5.56}.$$

Quick Check – Why stopping at 4.28 is incomplete

If your scratch work produced 4.28 for $V(B_3)$, here is the missing piece:

- A common shortcut is

  $$V(B_3) = 0.85\big[5 + \gamma \cdot 0\big] + 0.15\big[0.2 + \gamma \cdot 0\big] = 4.28,$$

  which treats *both* future-value terms as zero.

- That is correct for **Dig** (it terminates at B_4 where $V = 0$) but incorrect for **Nap**. Nap leaves Molebert in the *same* state, so its future value is $\gamma V(B_3)$, not 0.

- Incorporating that recursion yields

  $$V(B_3) = 4.28 + 0.1125\,V(B_3) \implies V(B_3) \approx 4.82.$$

Take-home rule: Whenever an action keeps the agent in the same state, the Bellman equation must include that state's value on both sides. Omitting the recursive term discards an infinite discounted tail of future reward—precisely what the Bellman framework is designed to capture.

2 Question 2 - Reinforcement Learning

Part 1. Model-Free vs. Model-Based RL (MCMA)

Scenario: In an automated greenhouse, two robots adjust watering schedules to maximize plant health:

- **Bot A** uses a model-free method. It directly tries watering levels, observes the resulting plant growth, and updates its policy purely from experience.
- **Bot B** is model-based. It learns a predictive model of how soil moisture, sunlight, and weather affect plant health, then plans actions using this model.

Which of the following statements about these bots are true? Select all that apply.

A. Bot B can plan future actions without taking real watering actions every time.

B. Bot A learns by predicting the effects of actions without trying them in the real greenhouse.

C. Bot B can suffer poor performance if its learned model is inaccurate.

D. Bot A will always be more sample efficient than Bot B, because it only needs real data.

E. Bot B needs more computational resources than Bot A to run planning or simulated rollouts.

F. Bot B's model is specialized and cannot be reused for different goals.

Correct Answers: A, C

Part 2. SARSA vs. Q-Learning (MCMA)

Scenario: A delivery drone must learn to fly through a forest to deliver packages. It can choose to fly **low** (fast but risky) or **high** (safe but slower).

- **Drone 1** uses **SARSA** (on-policy learning).
- **Drone 2** uses **Q-Learning** (off-policy learning).

Both drones use an ϵ-greedy exploration strategy during training.

Which of the following statements about how these drones learn are true? Select all that apply.

A. **Drone 1's Q-values depend on the actions it actually takes during learning, including its exploration behavior.**
True. SARSA updates follow the actions sampled by the policy, so the learned Q-values reflect real exploration.

B. **Drone 2 updates its Q-values using the maximum Q-value of possible next actions, regardless of what it actually does next.**
True. Q-Learning uses the max over actions in the next state, making it off-policy.

C. **SARSA's on-policy updates can make Drone 1 more conservative in risky areas compared to Drone 2.**
True. On-policy updates reflect the actual behavior, which can prefer safer paths when risky actions have bad expected returns.

D. **Drone 2's updates assume that it will always act optimally in the future, even if it sometimes explores.**
True. Q-Learning optimistically assumes that the next action is the best possible, not the actual exploratory action.

E. **Drone 1's Q-values represent the return under the optimal policy, not the behavior policy.**
False. SARSA's Q-values reflect the return under the current policy, which includes exploration.

F. **Both drones are guaranteed to learn the same final policy, regardless of their exploration behavior.**
False. Differences in exploration can lead to different policies, especially in stochastic or complex environments.

Correct Answers: A, B, C, D

Part 3. Greedy Exploration for Q-Learning

A factory uses reinforcement learning to choose maintenance actions for sensitive electronic components that degrade over time. At a given inspection, the system's Q-value estimates for each maintenance action are:

- Q(Full Replacement) = 8.0,
- Q(Partial Repair) = 7.5,
- Q(Increase Cooling) = 8.0,
- Q(Do Nothing) = 5.0.

The maintenance policy uses an ϵ-greedy approach with $\epsilon = 0.2$.

a) **Which actions are greedy?** Identify all actions that have the maximum Q-value.

b) **Probability of choosing Full Replacement** Compute the probability that Full Replacement is selected under this policy.

c) **Probability of choosing Increase Cooling** Compute the probability that Increase Cooling is selected under the same policy.

d) **Probability of choosing Do Nothing** Compute the probability that Do Nothing is chosen.

e) **Suppose the facility increases ϵ to 0.6. What is the new probability of choosing Increase Cooling?** Show your calculation.

f) **Explain the risk of setting ϵ too high in this context.** Why might it be risky for component reliability and electricity usage?

Solutions: Greedy Exploration for Q-Learning (Advanced)

a) **Greedy actions:** Full Replacement and Increase Cooling both have the highest Q-value (8.0).

b) **Probability of Full Replacement:** Two actions are greedy:

$$P(\text{greedy}) = 1 - \epsilon = 0.8. \quad P(\text{Full Replacement}) = \frac{0.8}{2} + \frac{0.2}{4} = 0.4 + 0.05 = 0.45.$$

c) **Probability of Increase Cooling:** Same as Full Replacement:

$$P(\text{Increase Cooling}) = 0.45.$$

d) **Probability of Do Nothing:** It's non-greedy:

$$P(\text{Do Nothing}) = \frac{\epsilon}{N} = \frac{0.2}{4} = 0.05.$$

e) **If** $\epsilon = 0.6$:

$$P(\text{Increase Cooling}) = \frac{0.4}{2} + \frac{0.6}{4} = 0.2 + 0.15 = 0.35.$$

f) **Risk of high exploration:** When ϵ is too high, the system picks suboptimal actions more often. This can lead to:

- More frequent random "Do Nothing" decisions, increasing the chance of unexpected component failures.
- Wasted electricity if unnecessary cooling is chosen too often.
- Overall instability in maintenance plans, harming production reliability and energy efficiency.

3 Question 3 - Game Theory

Part 1. Folk Theorem vs. Zero-Sum (MCMA)

Scenario: Two AI-controlled trading bots, Ada and Byron, repeatedly negotiate prices in an online commodities market. Each round, they can either **Collude** (keep prices high and share the market) or **Undercut** (lower prices aggressively to gain short-term market share).

- If both bots collude, they each earn steady profits over time.
- If one undercuts while the other colludes, the undercutter gains a larger immediate profit while the other loses market share.
- If both undercut, they drive prices down, reducing profits for both.

This negotiation repeats indefinitely, with each bot slightly discounting future profits. Each bot can retaliate against betrayal by undercutting in future rounds.

Which of the following statements about this scenario are true? Select all that apply.

A. **This is an example of a repeated game where the Folk Theorem applies.**
True. The Folk Theorem says that in infinitely repeated games, cooperation can be sustained by credible threats of punishment.

B. **Undercutting once will always lead to maximum profit in the long run.**
False. Undercutting can trigger retaliation, lowering long-term profits.

C. **A credible punishment strategy can make collusion stable.**
True. Bots can use threats of future undercutting to deter betrayal.

D. **This game is strictly zero-sum because the gain of one bot equals the loss of the other.**
False. Both can benefit by colluding, so total payoffs are not fixed.

E. **In a strictly zero-sum game, the Folk Theorem does not apply.**
True. The Folk Theorem applies to repeated games with non-zero-sum payoffs.

F. **If both bots discount the future heavily, collusion is more likely to hold.**
False. If future payoffs matter less, the incentive to maintain collusion weakens.

Correct Answers: A, C, E

Part 2. Mixed Strategies and Subgame Perfection (MCMA)

Scenario: Huey and Dewey are two robot pirates dividing loot on an island. They play a two-stage game:

- In the first stage, they simultaneously choose to either **Share Fairly** or **Steal**.

  - If both share, they split the treasure 50/50.
  - If one steals while the other shares, the stealer takes all.
  - If both steal, they fight, damaging each other and losing some loot.

- If they both choose to steal, they enter a subgame: they can either **Fight Hard** (risk higher damage for more loot) or **Back Down** (avoid damage but accept a smaller share).

Which of the following statements are true? Select all that apply.

A. **Huey can use a mixed strategy in the first stage to make Dewey indifferent between sharing and stealing.**
True. Randomizing can equalize Dewey's payoffs across actions.

B. **The second-stage fight can be analyzed for subgame perfect equilibrium.**
True. Each possible subgame must be optimal given earlier choices.

C. **A pure strategy will always yield better payoffs than mixing.**
False. Mixed strategies can be best responses when there's no pure-strategy Nash.

D. **A credible threat to fight in the subgame can influence first-stage choices.**
True. Backward induction means subgame threats affect earlier moves.

E. **This pirate game is strictly zero-sum because one pirate's gain equals the other's loss in every outcome.**
False. When both share fairly, they create positive sum payoffs.

F. **Subgame perfection requires randomizing in every subgame.**
False. Not every subgame needs mixed strategies; some may have pure equilibria.

Correct Answers: A, B, D

Part 3. Von Neumann Minimax with Mixed Strategy (Numerical)

Huey, Dewey, and Louie are selling **saltwater taffy** on a crowded boardwalk. Huey must pick one of 6 stand locations (A–F). Dewey and Louie adjust their stand signs and promotions to **minimize Huey's daily profit**.

Huey wants to maximize his worst-case profit but considers a mixed strategy to stay unpredictable.

Option	**R1**	$	**R2**	$	**R3**	$	**R4**	$
A	$R1$	\$950	$R2$	\$800	$R3$	\$450	$R4$	\$200
B	$R1$	\$900	$R2$	\$700	$R3$	\$600	$R4$	\$400
C	$R1$	\$1000	$R2$	\$500	$R3$	\$500	$R4$	\$350
D	$R1$	\$850	$R2$	\$850	$R3$	\$750	$R4$	\$600
E	$R1$	\$1200	$R2$	\$700	$R3$	\$400	$R4$	\$250
F	$R1$	\$1100	$R2$	\$800	$R3$	\$800	$R4$	\$700

Questions

a) Compute the worst-case daily profit for each stand option.

b) Which stand should Huey choose under the pure minimax strategy?

c) Suppose Huey uses a mixed strategy: 40% at Spot F and 60% at Spot D. Compute the **expected worst-case daily profit** for this mixed strategy.

d) Does this mixed strategy give a higher worst-case expected profit than the pure minimax option? **(Answer: Yes or No)**.

Solution: Von Neumann Minimax with Mixed Strategy (Big Numbers)

a)

$$\begin{aligned}
\min(A) &= \$200 \\
\min(B) &= \$400 \\
\min(C) &= \$350 \\
\min(D) &= \$600 \\
\min(E) &= \$250 \\
\min(F) &= \$700
\end{aligned}$$

b)

$$\max\{\$200, \$400, \$350, \$600, \$250, \$700\} = \$700 \Longrightarrow \text{Pure minimax} = \text{Spot F}.$$

c) Mixed strategy: 40% Spot F (worst = \$700) + 60% Spot D (worst = \$600)

$$\text{Expected worst-case} = 0.4 \times 700 + 0.6 \times 600 = 280 + 360 = 640.$$

d)

$$640 < 700 \Longrightarrow \text{No, the mixed strategy does not improve the worst-case.}$$

4 Question 4 - Game Theory Continued

Part 1. Folk Theorem and Subgame Perfection (MCMA)

Scenario: A fleet of autonomous robots in a furniture factory builds custom chairs for many clients. To stay productive, robots must share special cutting tools and upholstery machines fairly. Each robot can choose to **Cooperate** (wait its turn and share) or **Defect** (jump the queue, hoard tools). They interact indefinitely, monitor each other's actions, and can punish selfish behavior.

Which of the following statements about this scenario are true? Select all that apply.

A. **If robots value future production highly enough, they will tolerate short-term losses to keep cooperation stable.**
Correct. High discount factors make future rewards outweigh the temptation to defect once.

B. **A robot fleet can use a Grim Trigger strategy: if any robot cuts the line once, others refuse to share with it forever.**
Correct. Threat of permanent punishment deters cheating.

C. **A Nash equilibrium in this repeated setting means robots always share tools fairly in every round.**
Incorrect. Not all Nash equilibria are cooperative; some allow defection if no credible punishment exists.

D. **Subgame perfection ensures that no robot can bluff about punishing defectors if it wouldn't really do so later.**
Correct. SPE rules out empty threats that wouldn't be credible in practice.

E. **The Folk Theorem implies many ways of sharing tools fairly can be sustained as equilibria, not just one.**
Correct. Any feasible, individually rational payoff can be enforced with credible threats.

F. **Robots can use threats like blocking access to tools in future production rounds to enforce fair sharing.**
Correct. Future punishment is the practical mechanism that makes the Folk Theorem work.

Correct Answers: A, B, D, E, F

Part 2. Pavlov, Tit-for-Tat, and Zero-Sum with Puddles (MCMA)

Scenario: Huey, Dewey, Louie, and Puddles are Mars rovers exploring and sending data to Earth. They must share scarce solar charging pads and limited bandwidth. Each rover can **Cooperate** (take turns using resources) or **Defect** (hog them).

- Huey uses a **Tit-for-Tat** strategy: cooperate first, then mirror the last action.
- Louie uses a **Pavlov** (Win-Stay, Lose-Shift) strategy: repeat the last action if the payoff was good, switch if cheated.

- Dewey uses a **Grim Trigger**: always cooperate until defected against, then defect forever.
- Puddles plays selfishly: always defects and tries to take more than its share.

Sometimes the rovers also enter direct zero-sum contests for exclusive data slots. Puddles never believes in cooperation and ignores punishment.

Which of the following statements about this scenario are true? Select all that apply.

A. **Puddles's always-defect strategy can destabilize the cooperation assumptions for Huey, Dewey, and Louie.**
Correct. Persistent defection breaks the trust that Grim Trigger, Pavlov, and Tit-for-Tat rely on.

B. **Louie's Pavlov strategy may get trapped in constant switching against Puddles's selfishness.**
Correct. Win-Stay, Lose-Shift can get caught in loops if there's no reciprocation.

C. **Huey's Tit-for-Tat punishes Puddles each time it defects, but if Puddles never cooperates, the punishment has no effect.**
Correct. Tit-for-Tat relies on the other side wanting to return to cooperation.

D. **Dewey's Grim Trigger means once Puddles defects, Dewey will never cooperate again with Puddles.**
Correct. Grim Trigger is unforgiving and aims to deter defection by threatening eternal punishment.

E. **The Folk Theorem guarantees all four rovers can maintain stable cooperation no matter how Puddles behaves.**
Incorrect. Folk Theorem depends on credible punishment; if a player ignores punishment, it may break down.

F. **The other rovers may have to form an implicit alliance to punish Puddles in the zero-sum contests.**
Correct. Coalition behavior can limit a selfish player's gains.

Correct Answers: A, B, C, D, F

Part 3. Pythagoruff Donut Heist

Your mischievous dog, **Pythagoruff**, loves stealing donuts off your kitchen table at night. To discourage this, you use a spray bottle as punishment.

Actions per round:

- You: **Ignore (I)** or **Guard (G)**
- Pythagoruff: **Steal (S)** or **Stay (T)**

Updated Payoffs (per round):

- Ignore + Steal: You = -4, Pythagoruff = +3
- Ignore + Stay: You = +2, Pythagoruff = 0
- Guard + Steal: You = -1, Pythagoruff = -3
- Guard + Stay: You = -1, Pythagoruff = 0

3-Round Sequence:

1. Round 1: Pythagoruff Steals, You Guard
2. Round 2: Pythagoruff Steals, You Guard
3. Round 3: Pythagoruff Steals, You Ignore *(Policy Switch)*

Questions:

a) Compute your total payoff over 3 rounds if you switch to Ignore in Round 3.

b) Compute Pythagoruff's total payoff over 3 rounds if you switch.

c) Compute your total payoff if you did **not switch** in Round 3 (Guard instead).

d) Compute Pythagoruff's total payoff if you did not switch.

e) Compute the difference in your total payoff between switching and not switching.

f) Based on your total payoffs, should you have switched strategies in Round 3? (Yes or no.)

Solution: Pythagoruff Donut Heist

a) Your Total Payoff (Switch)

$$\begin{aligned}
\text{Round 1:} \quad & G + S = -1 \\
\text{Round 2:} \quad & G + S = -1 \\
\text{Round 3:} \quad & I + S = -4 \\
\implies \quad & \text{Total} = -1 + (-1) + (-4) = \boxed{-6}.
\end{aligned}$$

b) Pythagoruff's Total Payoff (Switch)

$$\begin{aligned}
\text{Round 1:} \quad & G + S = -3 \\
\text{Round 2:} \quad & G + S = -3 \\
\text{Round 3:} \quad & I + S = +3 \\
\implies \quad & \text{Total} = -3 + (-3) + 3 = \boxed{-3}.
\end{aligned}$$

c) Your Total Payoff (No Switch)

$$\begin{aligned}
\text{Round 3:} \quad & G + S = -1 \\
\implies \quad & \text{Total} = -1 + (-1) + (-1) = \boxed{-3}.
\end{aligned}$$

d) Pythagoruff's Total Payoff (No Switch)

$$\begin{aligned}
\text{Round 3:} \quad & G + S = -3 \\
\implies \quad & \text{Total} = -3 + (-3) + (-3) = \boxed{-9}.
\end{aligned}$$

e) Payoff Difference

Your Payoff (Switch) $= -6$, Your Payoff (No Switch) $= -3$. Difference $= -6 - (-3) = \boxed{-3}$.

f) Should You Have Switched?

Numerically, sticking with the original policy (Guard) yields a total of -3 vs. -6 if you switched.

$$\implies \text{No, you should not have switched strategies.} \quad \boxed{-3 > -6}$$

By staying consistent, your losses were minimized, and Pythagoruff's payoff was kept more negative.